

Lecture Plan Manufacturing Process (NMES101)

Course Type	Course Code	Name of Course	L	T	P	Credit
SEC2	NMES101	Manufacturing Process	2	0	3	3.5

Course Objective

In this course, the students will be exposed to the common manufacturing processes such as casting, forming, machining, welding as well as advanced manufacturing processes. Additionally, metrological principles, quality control practices, and computer control technology of machines will also be introduced. Further, the practical assignments of this course consist of demonstration and hands-on experience of these processes to make students confident to handle projects in future.

Learning Outcomes

Upon successful completion of this course, students will:

- have understanding of the safety rules, mechanical measurements and standardization.
- have understanding various types of casting processes for the industrial application.
- be able to acquire good knowledge metal working/forming processes.
- be able to acquire good knowledge of various types of machines and machining processes.
- have knowledge of metal sheet/plate cutting and welding processes.
- have knowledge of advanced manufacturing processes.

Unit No.	Topics to be Covered	L+P Hours	Learning Outcome
1	(i) Introduction and general safety rules for the workshop, principle of metrology. Standardization - concepts and National/International Standards, and relevant practical. Accuracy Precision and Calibration (ii) Introduction to commonly used mechanical devices, gauges/dial test indicators, CMM, Surface roughness measuring instrument and other shop floor measuring devices,	4 + 6	Knowledge of safety rules in the workshop, standardization and various types of measuring instruments and hands-on-practice on some measuring instruments. Prof. N. K. Singh (2 hr) Prof. Omkar Mypati (2 hr)
2	Patterns and allowances, Pattern design and materials, Pattern allowances and numerical, Riser design and numerical, Sand mould casting, Cast materials, Solidification time and numerical, Casting defects and remedies, Introduction to other casting processes and relevant practical.	4 + 6	Theoretical knowledge of casting methodology and design practice, and hands-on-practice on sand mould casting process. Prof. S. Saha (4 hr)
3	Principle of metal forming, hot and cold working processes, rolling, forging, extrusion, drawing, bulk deformation and sheet deformation processes, tube manufacturing processes, sheet metal working principle and applications and relevant practical.	4 + 6	Theoretical knowledge of various metal forming/working processes and hands-on-practice of making sheet metal tray. Prof. S. Chattopadhyaya (4 hr)
4	Principle of material removal operation, Machine Tools- lathe, shaping, milling, drilling, boring and grinding machines, cutting tools, selection of speeds, feed, depth, machine drive and work holding	6 + 9	Theoretical knowledge of various machining processes and hands-on-practice lathe, shaper, milling, drilling, machines for making jobs. Prof. N. K. Singh (3 hr)

	devices, MCD, tool wear and life and relevant practical.		Prof. Omkar Mypati (3 hr)
5	Working principle of sheet metal/plate cutting through gas, plasma, lasers. Principles of gas and arc welding, brazing, and soldering. Solid state welding. Types of weld joints, welding defects and remedies and relevant practical.	4 + 6	Theoretical knowledge of various metal cutting and welding processes and hands-on-practice sheet/plate cutting and welding. Prof. S. Chattopadhyaya (4 hr)
6	(i) Principle of WAJM, USM, CHM, ECM, EDM, Laser machining. (ii) CNC operation, Surface engineering, deployment of robot in manufacturing processes.	6 + 9	Knowledge and understanding of advanced manufacturing processes. Prof. Omkar Mypati (4 hr) Prof. S. Saha (2 hr)

Text book:

1. Manufacturing Science, Ghosh and Mallick, East-West Press
2. Principles of Modern Manufacturing, M. P. Groover, Wiley India

Reference Books:

1. Workshop Technology Part I, II, III by W. A. J. Chapman, CBS Publishers
2. Manufacturing Engineering and Technology by S. Kalpakjian and Schmid, Pearson Education
3. Elements of Workshop Technology (I & II) by Hazra & Chaudhary, Media Promoters & Publishers